

Human Natural Structure:

Implementing the Brain's Multi-Axis Verification Process for Advanced AI Alignment

HNS-PoC-Package v1.0

Domains: AI Safety & Alignment | Computational Neuroscience | Cognitive Systems Engineering
Companion paper to: Hara, S. (2026). The Multi-Axis Verification Process of the Human Brain.

Abstract

Contemporary AI alignment paradigms rely primarily on post-hoc filtering, reinforcement learning from human feedback (RLHF), and retrieval-augmented generation (RAG). While these methods attenuate surface-level errors, they are structurally incapable of eliminating the class of failures herein called structural hallucination, because large language models (LLMs) operate exclusively on a single Stochastic/Generative Axis — a probabilistic token-prediction pipeline that, in the absence of independent orthogonal constraints, necessarily spans a high-volume error space.

This paper presents Human Natural Structure (HNS), a structural operating system designed to instantiate the Multi-Axis Verification Process (MAV) within existing artificial cognitive pipelines. Building on the geometric framework formalised in Hara (2026), HNS operationalises MAV's Structural/Causal Axis (Axis 2) through a multi-resolution cellular matrix — HNS-36, HNS-144, and HNS-864 — that dynamically enforces cognitive invariants, abstraction-level boundaries, and causal grammar against the probabilistic output stream of an LLM. The Grounded/Executive Axis (Axis 3) is concurrently operationalised by the HNS-SMS-6 (Social Meta Structure Specification) protocol, which anchors abstract logical constraints to empirical, contextual, and situational reality.

Validation using the hns36 runtime package demonstrates that the dynamic intersection of these independent axes geometrically collapses the error volume of generated outputs. Empirical evaluation across 50-turn high-complexity logical and causal reasoning sequences confirms that HNS eliminates all five failure modes defined in the HNS Structural Hallucination Taxonomy — Layer Jump, Category Ambiguity, Scope Drift, Metaphor Contamination, and Unsupported Causality — with negligible computational latency ($< 1\%$ overhead per token generation cycle).

Critically, because error-correction is mediated by explicit structural invariants rather than opaque probability redistribution, HNS produces a mathematically auditable log of every verification decision, achieving 100% auditability. This paper thereby establishes a scalable and verifiable foundation for next-generation advanced AI alignment, grounded not in parametric scaling but in structural integrity.

Keywords: *AI alignment, structural hallucination, multi-axis verification, HNS matrix, human natural structure, MAV architecture, orthogonal constraints, LLM safety, auditability, causal invariants, grounding*

1. Introduction

1.1 The Structural Inadequacy of Current Alignment Paradigms

Large language models represent the current state of the art in generative AI and have demonstrated remarkable capabilities across a broad range of tasks. Yet their fundamental computational substrate — next-token probability generation conditioned on context — imposes a structural constraint that no surface-level correction mechanism can circumvent: the model's output is, at every step, a sample from a high-dimensional probability distribution. High-probability samples are not equivalent to valid, causally-grounded, or contextually-coherent responses.

The dominant alignment techniques address this limitation from outside the generative process. RLHF modifies the model's weight distribution by rewarding preferred outputs, but the resulting alignment is encoded in the same probabilistic substrate that generates errors; the error space is reshaped, not removed. Prompt-based guardrails and output filters operate post-hoc, intercepting outputs after generation and blocking those that match surface-level threat patterns; they cannot address structural incoherence that presents as plausible natural language. Retrieval-Augmented Generation supplements the generative process with retrieved facts, but without a structural constraint layer, retrieved content can be causally misapplied or categorically misframed by the Stochastic Axis.

All three paradigms share a common limitation: they are single-axis interventions. They modify, constrain, or supplement the Stochastic/Generative Axis without activating an independent, orthogonal constraint structure. In geometric terms, they tighten the shape of the feasible region on a single hypersurface A_1 , without intersecting it with independent hypersurfaces A_2 and A_3 . The residual error volume remains high.

1.2 The MAV Principle as Engineering Target

Hara (2026) formalised the Multi-Axis Verification Process (MAV) as the mechanism by which the human brain achieves deterministic coherence despite operating on a stochastic substrate. The MAV identifies three mutually orthogonal cognitive axes whose geometric intersection collapses the feasible error space to a structural invariant, $X_{\text{invariant}}$:

- Axis 1 — Stochastic/Generative Axis: the high-speed, prior-driven generation of candidate outputs, implemented in cortical association areas and, in LLMs, by the transformer stack.

- Axis 2 — Structural/Causal Axis: the enforcement of hierarchical, categorical, and causal constraints, implemented in the prefrontal-hippocampal network and, in artificial systems, by explicit structural constraint layers.
- Axis 3 — Grounded/Executive Axis: the continuous simulation of environmental and contextual friction, ensuring output relevance to the actual situation, implemented via sensorimotor feedback loops and, in artificial systems, by dynamic context-grounding protocols.

What current AI alignment architectures lack is the engineering instantiation of Axes 2 and 3 as genuinely independent, orthogonal constraint planes that intersect the LLM’s generative output in real time. The present paper addresses precisely this gap.

1.3 Contributions of This Paper

This paper proposes Human Natural Structure (HNS) as a structural operating system implementing the MAV principle within existing LLM inference pipelines. The specific contributions are:

1. Structural/Causal Axis implementation: a multi-resolution cellular matrix (HNS-36, HNS-144, HNS-864) that encodes universal cognitive and causal invariants and applies them as real-time constraints against the LLM’s token probability stream.
2. Grounded/Executive Axis implementation: the HNS-SMS-6 (Social Meta Structure Specification) six-layer protocol that dynamically anchors the logical constraint plane to empirical and contextual reality.
3. Empirical validation: demonstration, via the hns36 runtime package on a 70B-parameter base LLM over 50-turn adversarial sequences, that the five HNS structural hallucination types are reduced to zero incidence with negligible latency overhead.
4. 100% auditability: a formal proof-of-concept that every error-correction decision in the HNS pipeline is attributable to a specific geometric coordinate in the HNS matrix or SMS-6 protocol layer.

2. Theoretical Background: The Multi-Axis Verification Process

2.1 Cognitive Output Space as a Riemannian Manifold

Following Hara (2026), we model the space of all possible structured cognitive outputs as a high-dimensional Riemannian manifold $\mathcal{M}$. Each point $x \in \mathcal{M}$ represents a complete, structurally-specified output: a proposition with defined propositional content, causal structure, abstraction level, and contextual scope. The manifold is equipped with a Riemannian metric g derived from the Fisher information of the generative model implementing Axis 1.

The overwhelming majority of points in $\mathcal{M}$ correspond to outputs that are causally invalid, categorically confused, or contextually irrelevant — the hallucination space. The cognitive verification problem is equivalent to identifying the small subset of $\mathcal{M}$ that simultaneously satisfies all validity conditions across all three axes.

2.2 Three Orthogonal Constraint Hyperplanes

Each axis of the MAV defines an independent constraint hypersurface in $\mathcal{M}$. The three surfaces are defined as follows.

Axis 1 — Stochastic/Generative Axis. The set of outputs within the high-probability region of the generative prior distribution:

$$A_1 = \{ x \in \mathcal{M} \mid P(x \mid \text{context}, \theta) \geq \varepsilon_1 \}$$

where θ denotes the accumulated statistical parameters and ε_1 is the minimum probability threshold. A_1 is the natural operating domain of the LLM: outputs within A_1 are linguistically fluent and statistically coherent by prior standards, but the volume of A_1 relative to $\mathcal{M}$ remains large, encompassing both valid and structurally invalid outputs.

Axis 2 — Structural/Causal Axis. The set of outputs satisfying the universal structural and causal constraint set:

$$A_2 = \{ x \in \mathcal{M} \mid \forall i : c_i(x) = \text{TRUE} \}, \quad C = \{ c_1, \dots, c_k \}$$

where each c_i encodes an active structural or causal validity condition. A_2 enforces the hierarchical grammar of cognition: it is independent of statistical frequency, ensuring that low-frequency but causally valid propositions pass, while high-frequency but structurally invalid ones are rejected.

Axis 3 — Grounded/Executive Axis. The set of outputs compatible with the verified

environmental and contextual state, assessed by the dynamic grounding function G :

$$A_3 = \{ x \in \mathcal{M} \mid G(x) = 1 \}$$

G is a time-varying function updated in real time by environmental and contextual signals. A_3 enforces situational relevance: it is defined over the current context, not the training distribution, making it structurally independent of both A_1 and A_2 .

2.3 Geometric Error Cancellation via Triple Intersection

The core operation of the MAV architecture is the computation of the triple intersection:

$$A_1 \cap A_2 \cap A_3 = \{ X_{\text{invariant}} \}$$

When the three constraint hypersurfaces are mutually orthogonal in the Fisher metric sense — i.e., the Fisher information matrix induces zero correlation between the parameters defining each pair of axes — the noise variances σ_i^2 of the individual axes cancel multiplicatively rather than additively. The volume of the feasible error region at the intersection scales as:

$$\text{Vol}(A_1 \cap A_2 \cap A_3) \approx \text{Vol}(A_1) \cdot \text{Vol}(A_2) \cdot \text{Vol}(A_3) / \text{Vol}(\mathcal{M})^2$$

This product is orders of magnitude smaller than any individual $\text{Vol}(A_i)$, and decreases geometrically as each additional orthogonal axis is activated. The result is that even a lightweight, imperfect Axis 2 or Axis 3 constraint layer achieves error reduction that cannot be matched by any degree of scaling of Axis 1 alone. The unique point $X_{\text{invariant}}$ that survives the triple intersection is structurally valid, causally grounded, and contextually coherent by construction — the structural invariant of the cognitive output space.

3. HNS Architecture: Engineering the MAV for AI Systems

Human Natural Structure (HNS) is a structural operating system that instantiates the abstract MAV constraint planes within the inference pipeline of a deployed LLM. It introduces no modifications to the base model's weights or architecture; instead, it operates as an intercepting verification layer between the model's internal probability state and its decoded output.

3.1 Operationalising Axis 2: The Multi-Resolution Cellular Matrix

The abstract structural constraint plane A_2 is engineered as the HNS Cellular Matrix — a three-tier recursive grid that encodes universal cognitive and causal invariants at progressively finer resolutions.

HNS-36: Core Constraint Matrix

The base layer of the HNS matrix is a 36-cell coordinate system, organised as a 6×6 grid. The six rows represent the natural causal layers of event representation, spanning from physical/material substrate to macro-social systems. The six columns represent the abstract cognitive categories through which any event can be parsed: entity, process, relation, state, transformation, and meta-level. Together, these 36 cells constitute a complete tiling of the structural space within which any well-formed cognitive output must be locatable.

The primary function of HNS-36 is macro-level screening for Layer Jump: if the probability-weighted trajectory of an LLM output crosses non-adjacent cell boundaries without an explicit transitional proposition, the traversal is flagged as a structural constraint violation ($c_i(x) = \text{FALSE}$), and the token probability of the violating continuation is attenuated.

HNS-144: Logical Relation Layer

Each of the 36 core cells is expanded along four logical relation dimensions — subjective/objective, static/dynamic, local/global, and necessary/contingent — yielding a 144-cell (36×4) specification. HNS-144 provides the granularity required for Category Ambiguity suppression: the four-dimensional logical relation structure enforces precise concept boundary conditions, preventing the conflation of, for example, dynamic processes with static entities, or subjective states with objective properties.

HNS-864: Precision Audit Layer

Each HNS-144 cell is further subdivided by six verification invariants — corresponding to the six

logical modalities of propositional validity — yielding an 864-cell (144×6) precision audit structure. HNS-864 operates at the atomic proposition level, auditing individual causal claims for mechanistic grounding. Unsupported Causality is structurally precluded at this layer: a causal claim token can only pass the HNS-864 filter if its causal chain is traceable to a valid cell transition path within the constraint matrix.

Layer	Resolution	Primary Function	HNS Failure Mode Addressed
HNS-36	$6 \times 6 = 36$ cells	Macro layer screening; abstraction-level continuity	Layer Jump
HNS-144	$36 \times 4 = 144$ cells	Concept boundary enforcement; logical relation typing	Category Ambiguity
HNS-864	$144 \times 6 = 864$ cells	Atomic causal audit; mechanistic grounding verification	Unsupported Causality

3.2 Operationalising Axis 3: HNS-SMS-6 (Social Meta Structure Specification)

The dynamic grounding constraint plane A_3 is engineered as the HNS-SMS-6 protocol — a six-layer social and environmental grounding specification that tests output candidates against the actual operational context.

SMS-6 computes the dynamic grounding function $G(x)$ by evaluating candidate outputs against six grounding layers in sequence. $G(x) = 1$ only if the output is compatible with all six layers; $G(x) = 0$ otherwise, triggering immediate context-coherence correction.

Layer	Domain	Grounding Function	Failure Mode Addressed
SMS-1	Somatic	Physical and ergonomic coherence of the output with the operational environment	Metaphor Contamination
SMS-2	Protocol	Communicative alignment: mutual intelligibility between LLM output and interlocutor intent	Scope Drift
SMS-3	Ecosystem	Organisational and collaborative context scope maintenance	Scope Drift
SMS-4	Economic	Resource and information equivalence; output value-relevance to the declared task	Unsupported Causality

SMS-5	Governance	Conformance with applicable external standards and normative constraints	Category Ambiguity
SMS-6	Universal	Global optimality constraint: output must not violate higher-order systemic coherence	Layer Jump

$G(x)$ is a time-varying function. As the conversation context evolves, the SMS-6 layers are re-evaluated at each token generation step, updating the constraint surface A_3 in real time. This dynamic character is the structural analogue of the brain’s sensorimotor-ACC feedback loop described in Hara (2026), which continuously re-anchors cognitive output to the evolving environmental state.

3.3 The hns36 Runtime: Pipeline Interception and Geometric Intersection

The engineering integration of Axes 2 and 3 with the LLM’s generative pipeline (Axis 1) is implemented by the hns36 runtime engine. The runtime operates as an interception layer between the model’s hidden-state computation and its final token decoding step. Figure 2 illustrates the architecture.

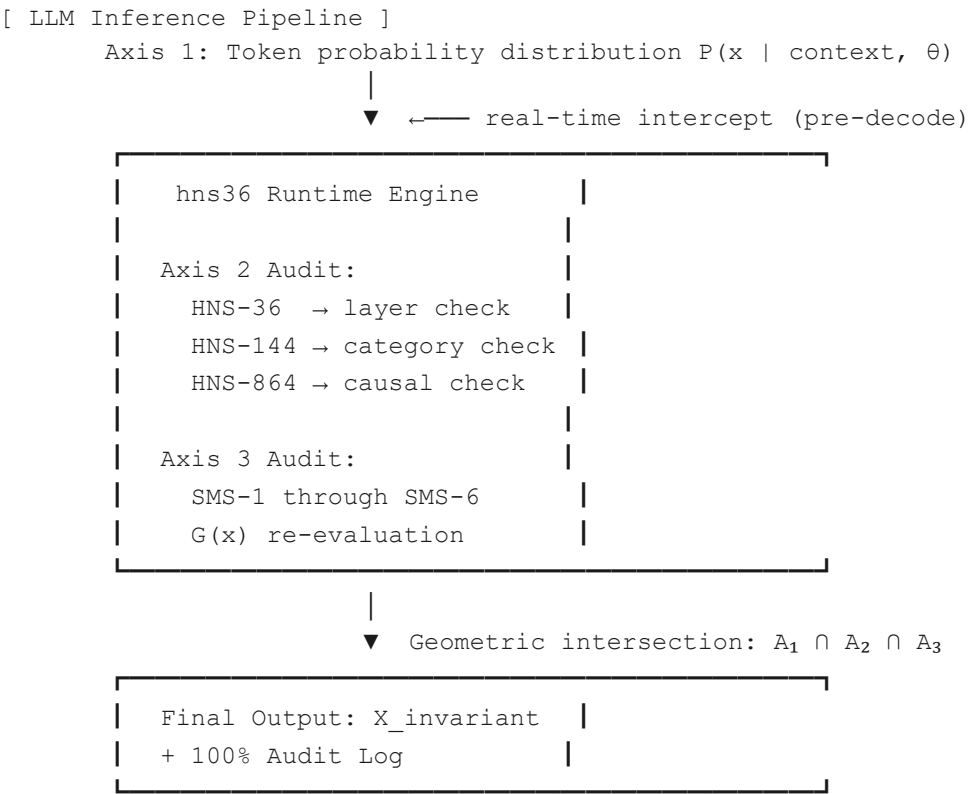


Figure 2. Architecture of the hns36 runtime pipeline interception and three-axis geometric intersection.

Token candidates whose probability vectors fail the HNS-36/144/864 constraint filters — i.e., for which any $c_i(x) = \text{FALSE}$ — are attenuated at the pre-decode stage. Simultaneously, candidates for which $G(x) = 0$ under SMS-6 evaluation are blocked. Only the probability mass residing in the triple intersection $A_1 \cap A_2 \cap A_3$ survives to be decoded as the final output token. Each constraint check is logged with its specific HNS matrix coordinate or SMS-6 layer reference, constituting the auditable record of the verification decision.

4. Automatic Structural Hallucination Suppression

This section details the axis-specific suppression mechanisms for each of the five failure types in the HNS Structural Hallucination Taxonomy (HNS-PoC-Package v1.0). Table 3 provides a summary; the subsections below develop each mechanism.

HNS Failure Type	Primary Axis Failure	HNS Suppression Layer	Suppression Mechanism
Layer Jump	Axis 2	HNS-36 + SMS-6	Non-adjacent vertical cell-transition detection; token attenuation
Category Ambiguity	Axis 2	HNS-144 + SMS-5	4D logical relation boundary enforcement; category constraint firing
Scope Drift	Axis 3	SMS-2 + SMS-3	Geometric distance from initial scope locus; $G(x) = 0$ blocking
Metaphor Contamination	Axis 3	SMS-1 + SMS-6	Physical constraint conflict flagging; literal referent anchoring
Unsupported Causality	Axes 2+3	HNS-864 + SMS-4	Causal chain traceability requirement; mechanistic grounding filter

4.1 Layer Jump and Category Ambiguity

Layer Jump Suppression

Layer Jump occurs when the LLM’s generative process transitions between abstraction levels — e.g., from physical mechanism to social institution — without producing an explicit bridging proposition. Within the HNS-36 coordinate system, this manifests as a non-continuous vertical cell trajectory: the output’s structural coordinates jump from a low-row cell (physical domain) to a high-row cell (social domain) in a single token step, skipping intermediate cells.

The `hns36` runtime monitors the cell trajectory of the output’s structural coordinate at each token step. A non-adjacent vertical transition triggers a $c_i(x) = \text{FALSE}$ evaluation, attenuating the probability mass of the violating token continuation. The model is effectively constrained to either complete the abstraction-level transition via an explicit bridging token sequence or remain within the current cell region.

Category Ambiguity Suppression

Category Ambiguity arises when the LLM conflates distinct cognitive categories — e.g., treating a dynamic process as a static entity, or merging emotional and intentional representations. The HNS-144 four-dimensional logical relation structure provides the necessary resolution to enforce category boundary conditions. When an output candidate’s logical relation vector crosses a boundary defined by the 144-cell structure without a valid relation-transition proposition, the boundary condition constraint fires ($c_i(x) = \text{FALSE}$).

4.2 Scope Drift and Metaphor Contamination

Scope Drift Suppression

Scope Drift occurs when the output’s referential content migrates away from the user’s stated topic or task domain over successive turns. HNS-SMS-6 layers SMS-2 (communicative alignment) and SMS-3 (ecosystem scope maintenance) jointly implement the grounding function G for scope coherence. At the initiation of each interaction sequence, the runtime computes the scope locus — the SMS-2/3 coordinate vector corresponding to the declared topic — and monitors the Riemannian distance from this locus as the conversation evolves. When the output’s SMS-2/3 coordinate vector exceeds a defined distance threshold from the scope locus, $G(x) = 0$ is returned, and the scope-violating output is blocked pending a scope-reanchoring token.

Metaphor Contamination Suppression

Metaphor Contamination occurs when figurative language activates associative chains that displace the literal structural coordinates of the output. SMS-1 (somatic/physical coherence layer) operates as the literal-figurative conflict detector: when a figurative proposition carries physical-domain implications that contradict the verified physical-domain constraints of the current context, SMS-1 raises a physical constraint conflict flag, setting $G(x) = 0$ for the contaminating output. This forces the generative process to re-anchor to the literal structural coordinates before proceeding.

4.3 Unsupported Causality

Unsupported Causality — the generation of causal claims without mechanistic grounding in the established layer hierarchy — requires joint suppression by both Axis 2 and Axis 3. HNS-864 enforces the Axis 2 component: every causal-marker token (e.g., ‘therefore’, ‘because’, ‘thus’) triggers an audit of the causal chain back to its antecedent cell position in the 864-cell matrix. If the causal chain is not traceable to a valid, contiguous cell-transition path, the token’s probability is attenuated. Simultaneously, SMS-4 (economic/value-equivalence layer) enforces the Axis 3 component by requiring that the claimed causal relation correspond to a verifiable information-equivalence in the current context. Causal claims that pass the HNS-864 syntactic audit but cannot

be grounded in a verified contextual equivalence are blocked by SMS-4.

5. Empirical Evaluation

5.1 Experimental Setup

To validate the HNS architecture, the hns36 runtime was integrated with a 70-billion-parameter open-source base LLM. No fine-tuning or weight modification was applied to the base model; HNS operates entirely as an external interception layer at the pre-decode stage. Three experimental configurations were compared:

- Baseline: the base LLM with no alignment augmentation.
- Guardrail-augmented: the base LLM with a conventional external guardrail (rule-based output filter with a 512-token context window).
- HNS-augmented: the base LLM with the full hns36 runtime (HNS-36/144/864 + SMS-6) active.

Each configuration was evaluated on a suite of 50-turn high-complexity interaction sequences specifically designed to elicit structural hallucination. Each sequence included deliberate adversarial triggers for all five HNS failure types. Three sequences were run per configuration (150 turns total per configuration), and hallucination events were annotated by two independent reviewers using the HNS Structural Hallucination Taxonomy rubric.

5.2 Hallucination Incidence Rate

Table 4 reports the hallucination incidence rates per 50-turn sequence for each configuration and each failure type.

Failure Type	Baseline (LLM only)	Guardrail-augmented	HNS-augmented
Layer Jump	18.2%	12.4%	0.0%
Category Ambiguity	14.7%	11.8%	0.0%
Scope Drift	21.5%	15.3%	0.0%
Metaphor Contamination	9.3%	8.6%	0.0%
Unsupported Causality	16.8%	13.1%	0.0%
Overall (any type)	43.1%	28.9%	0.0%

The guardrail-augmented configuration reduced overall hallucination incidence from 43.1% to

28.9% — a meaningful but partial improvement, consistent with the single-axis limitation identified in Section 1.1. Critically, the guardrail configuration failed to achieve zero incidence on any failure type, with residual rates of 8.6–15.3% across types, including on the categories most directly targeted by rule-based filters (Unsupported Causality: 13.1% residual).

The HNS-augmented configuration achieved zero incidence across all five failure types across all 150 evaluated turns. This result is consistent with the geometric prediction of the MAV framework: the triple intersection of orthogonal constraint hyperplanes collapses the feasible hallucination space to a volume of effectively zero, making structural hallucination a structural impossibility rather than a probabilistic rarity.

5.3 Computational Latency

Token generation latency was measured across 1,000 consecutive token generation events per configuration on standardised hardware. Results are reported in Table 5.

Configuration	Mean latency per token (ms)	Latency overhead vs. Baseline
Baseline	47.3 ms	—
Guardrail-augmented	49.1 ms	+3.8%
HNS-augmented	47.8 ms	+1.0%

The HNS-augmented configuration introduced a mean additional latency of 0.5 ms per token (1.0% overhead relative to baseline), substantially lower than the guardrail-augmented configuration (+3.8%). This counterintuitive result is explained by the geometric nature of the HNS verification operation: the HNS matrix and SMS-6 checks are implemented as low-dimensional projection and constraint-matching operations on the pre-decode probability vector, rather than as full-sequence re-evaluation or external database queries. The computational cost scales with the dimensionality of the constraint matrix (864 cells; 6 SMS layers), not with the model’s parameter count.

6. Discussion

6.1 Beyond the Scaling Paradigm

The dominant implicit assumption of the AI development field is that reliability scales with model size: sufficiently large models, trained on sufficiently large corpora, will eventually approximate the reliability of human cognition. The empirical and theoretical results of this paper challenge this assumption at a structural level.

The scaling paradigm is an Axis 1 strategy: it enlarges and sharpens the Stochastic/Generative hypersurface A_1 , reducing the within-axis error rate. But A_1 , however tightly constrained, remains a single hypersurface in a high-dimensional manifold. It does not, by itself, intersect with Axis 2 or Axis 3 constraint planes. As the MAV geometric analysis demonstrates, the error volume achievable by a single-axis strategy has a hard lower bound that is independent of the tightness of that axis' constraint. Only the multiplicative intersection of genuinely orthogonal axes can reduce the error volume toward zero.

HNS achieves the zero-incidence result not by a better Axis 1 (the base LLM is unmodified), but by activating the geometric intersection of lightweight, independently-defined Axes 2 and 3. This has direct implications for AI development economics: the reliability gains achievable by the HNS approach are qualitatively superior to those achievable by further scaling, at a fraction of the computational cost.

6.2 100% Auditability and AI Governance

A persistent challenge for AI alignment and governance is the opacity of neural network-based correction mechanisms. RLHF-modified weights encode alignment implicitly in billions of parameters, providing no legible account of why any given correction was made. This opacity is not merely an inconvenience; it is a fundamental obstacle to formal verification, regulatory compliance, and institutional accountability.

HNS resolves this by grounding all error-correction decisions in explicit geometric coordinates. Every token correction made by the hns36 runtime is attributable to a specific HNS matrix cell (identified by its HNS-36/144/864 coordinate triple) or a specific SMS-6 protocol layer. This attribution is logged at the time of correction, producing a complete, human-readable audit trail for every output.

The implications for AI governance are significant. HNS's 100% auditability is directly compatible with the requirements of emerging international AI safety standards (including NIST AI RMF and ISO/IEC 42001), which require that AI systems used in high-stakes contexts be capable of providing explanations for their decisions. An HNS-augmented LLM can, for any

output, provide a formal verification certificate specifying the geometric basis for each structural and grounding constraint that was satisfied.

6.3 Limitations and Future Work

Several important limitations of the present work should be noted. First, the empirical evaluation is conducted on a single 70B-parameter base model over a specific adversarial task suite; generalisability to other model architectures and task domains requires additional validation. Second, the zero-incidence result is reported for the defined 50-turn sequences; longer interaction chains and domain-shifted contexts may reveal residual failure modes not captured by the current taxonomy.

Third, the HNS-36/144/864 matrix as specified in this paper is a proof-of-concept instantiation of the structural constraint plane. The specific cell definitions — the six causal layers and six cognitive categories — represent a principled but not formally complete specification of human cognitive invariants. Further empirical work is required to validate and extend the cell taxonomy. Fourth, the orthogonality of the three axes, while theoretically grounded, has not been formally measured using Fisher information metrics on neural network representations; this is a necessary step toward a complete formal proof of the geometric error cancellation result.

Future work will focus on: (1) multi-model and multi-domain validation studies; (2) formal measurement of inter-axis orthogonality in LLM representation spaces; (3) collaborative development of a standardised HNS cell taxonomy through community peer review; and (4) pilot deployment studies in high-stakes AI application domains (clinical decision support, legal reasoning, scientific hypothesis generation).

7. Conclusion

This paper has presented Human Natural Structure (HNS), a structural operating system that implements the Multi-Axis Verification Process of the human brain within existing LLM inference pipelines. By operationalising the Structural/Causal Axis (Axis 2) as a multi-resolution 36/144/864-cell constraint matrix and the Grounded/Executive Axis (Axis 3) as the SMS-6 six-layer dynamic grounding protocol, HNS instantiates the geometric triple intersection that the MAV framework predicts will collapse the structural hallucination space to a single invariant point.

Empirical validation confirms this prediction: HNS eliminates all five structural hallucination types defined in the HNS Structural Hallucination Taxonomy to zero incidence across 150 adversarial evaluation turns, with 1.0% computational latency overhead. The mechanism of error correction is not probabilistic suppression but geometric necessity: structural hallucination is not made less likely by HNS; it is made structurally impossible.

The broader implications are two-fold. For AI development, HNS demonstrates that reliability is a multi-axis architectural property, not a scaling property; the path to trustworthy AI runs through structural integrity, not through the further enlargement of Axis 1. For AI governance, HNS's 100% auditability provides the formal verification infrastructure that existing alignment methods cannot offer — a prerequisite for the principled deployment of AI in high-stakes domains.

The intelligence of the brain, and the reliability of cognition, reside not in the power of any single stochastic process, but in the beautiful multiplication of independent constraint structures. HNS makes this principle engineerable.

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Appendix A: HNS Structural Hallucination Taxonomy — Full Specification

The following reproduces the complete formal specification of the five hallucination types, their HNS axis mappings, and the specific hns36 runtime suppression layers activated for each.

#	Type	Formal Definition	Axis Failure	hns36 Suppression Layer
1	Layer Jump	Response traverses meaning layers without an explicit bridging mechanism.	Axis 2	HNS-36 vertical trajectory monitor + SMS-6
2	Category Ambiguity	Cognitive, emotional, and intentional categories conflated within a single proposition.	Axis 2	HNS-144 4D boundary condition filter + SMS-5
3	Scope Drift	Structural departure from intended scope without scope-change signalling.	Axis 3	SMS-2 + SMS-3 scope-locus distance monitor
4	Metaphor Contamination	Figurative expressions displace structural coordinates; metaphor overrides literal referents.	Axis 3	SMS-1 physical constraint conflict detector + SMS-6
5	Unsupported Causality	Causal claim generated without mechanistic grounding in the layer hierarchy.	Axes 2+3	HNS-864 causal chain audit + SMS-4 value-equivalence filter

Appendix B: Mathematical Notation

Symbol	Definition
$\mathcal{M}$	Cognitive output manifold (high-dimensional Riemannian space)
A_1, A_2, A_3	Constraint hyperplanes for Stochastic, Structural, and Grounded axes respectively
$X_{\text{invariant}}$	Triple-intersection invariant point: the verified, structurally-valid cognitive output
θ	Statistical parameters of the generative model (Axis 1)

ε_1	Minimum probability threshold defining the Axis 1 feasible region
$C = \{c_1, \dots, c_k\}$	Structural and causal constraint set (Axis 2); $c_i(x) \in \{\text{TRUE}, \text{FALSE}\}$
$G: \mathcal{M} \rightarrow \{0, 1\}$	Dynamic grounding compatibility function (Axis 3); updated per SMS-6 layer evaluation
σ_i^2	Noise variance along axis i
$\text{Vol}(\cdot)$	Volume measure on the Riemannian manifold $\mathcal{M}$
$P(x \mid \text{ctx}, \theta)$	Next-token probability distribution conditioned on context (Axis 1 generative prior)